

Middleware Lab: RabbitMQ Exchange Routing Architecture

Duration : 60 Min.

Scenario

A high-performance system needs to distribute messages selectively to different queues based on request routing keys.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Configure routing architectures using Direct, Fanout, and Topic exchanges with Dead Letter handling.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.

Validation

Once you have saved your files and are ready to submit, return to the platform dashboard and click the **"Submit Final"** button. This task is evaluated manually by the grading team.

Evaluation Rubric:

1. Defining Direct, Fanout, and Topic exchange routing paradigms: 5 marks
2. Configuring custom routing keys and queue bindings: 5 marks
3. Setting up Dead Letter Exchange (DLX) parameters for failed tasks: 5 marks
4. Validating message consumption flow across queues: 5 marks

Total Score: 20 Marks

Important Notes

- This is a manually evaluated question.
- Ensure all logs and configurations are fully saved in your workspace folder before submitting.